Tax Evasion and Productivity

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Tax evasion through input overreporting

- VAT allows firms to deduct taxes paid on their purchases from the taxes collected on sales
- By overreporting their inputs, firms can reduce their tax bill
- Input overreporting is widespread: a common practice and a concern among OECD countries (OECD 2017)
  - Losses of 1.2–2.7% of total tax revenue in Chile, Colombia, Ecuador, Poland and Slovakia
  - Conditional on detected evasion: true losses are plausibly larger
- Governments have limited resources, and overreporting costs the government tax revenue

Question: Can we detect and measure tax evasion through input overreporting? How much does overreporting of inputs cost the government?

Why is it difficult to detect?

- Overreporting is unobserved: firms conceal their actions because it is illegal
- Tax-evading firms might *pass* as low-productivity firms
- For a given level of output, high input use by a firm might be because the firm is overreporting or because its productivity is low

What have people done?

- **Transaction-level data and administrative records** (Carrillo et al. 2022; Zumaya et al. 2021)
  - False invoicing, ghost-firm invoices, invoice mills
  - Access to this data is restricted; it is conditional on detected evasion
- **On the consumer side:** a benchmark group to measure misreporting from potentially evading individuals (Pissarides and Weber 1989; Gorodnichenko et al. 2009)
  - Self-employed vs. employees
  - The self-employed might underreport their income to pay less income tax; employees cannot misreport because their income is reported by their employers
  - Consumption is a function of true income: comparing their consumption backs out how much the self-employed underreport

The natural analogue for firms: output

- Output is a function of true inputs (assuming output is correctly reported)
- If we knew the production function (PF) and productivity, we could back out true inputs: the difference between reported and true inputs is the overreporting
- However, we rely on *reported inputs* to recover the PF and productivity
  - Proxy-variable methods rely on the *flexible* input (Levinsohn and Petrin 2003; Ackerberg et al. 2015; Gandhi et al. 2020)
    - If the flexible input is deductible, firms might overreport it: materials, in my application
- We have to solve the problem simultaneously, for overreporting and productivity

What do I do?

- Instead of looking into the PF equation, I take the FOC of the firm's profit-maximization problem
- The FOC links the materials share of revenue to the materials elasticity of output, up to measurement error
  - It does not depend on productivity (Hicks-neutral productivity)
- Just like in the individuals' case, I estimate this elasticity from a benchmark group of firms that do not overreport
  - The benchmark group can have different productivity than the potentially tax-evading firms
  - The critical assumption: both share a common technology
- With the elasticity at hand, the difference between reported and true materials reveals the overreporting, up to measurement error

What do I find?

Colombian manufacturing, 1981–1991:

1. **Detect:** overreporting in 9 of the 18 industries liable for the sales tax (about two fifths of manufacturing output); none in the 2 exempt food industries
2. **Measure:** potentially tax-evading firms overreport on average 4.5–23.1% of their true materials, depending on the industry
3. **Cost:** undetected overreporting costs 4.7% of gross sales-tax revenue; a 1% increase in the rate on purchases lowers net sales-tax revenue by about 1.7% of gross sales-tax revenue
4. **Productivity:** failing to correct for overreporting leads to biased estimates of the PF and productivity

Setting: Colombia, 1981–1991

- Annual Survey of Manufacturing (EAM): a panel of manufacturing plants with more than 10 employees
  - Output and inputs, juridical organization, and the sales tax charged on sales and paid on purchases
- The Colombian sales tax worked as a VAT: manufacturers credited the tax paid on their purchases against the tax charged on their sales
- The credit rewards overreporting materials: each fictitious peso lowers the tax liability by the rate on purchases
  - Capital goods could not be credited and labour was not a taxed purchase
- Weak enforcement: tax evasion flourished, and the tax authority audited inefficiently (Sánchez Torres and Gutiérrez Sourdis 1994; Perry and Orozco de Triana 1990)
- Food products (311, 312) were exempt from the sales tax: almost no reward to overreporting

The benchmark group: corporations

- Two kinds of scrutiny restrained corporations from overreporting: government and market
  - Government: the Superintendent of Corporations closely supervised corporations and required each to have an auditor (Joint Tax Program of the Organization of American States and the Inter-American Development Bank 1965)
  - Market: tradable shares subject corporations to market discipline; the market penalizes aggressive tax behaviour even when it is legal (Hanlon and Slemrod 2009)
- Limited liability companies (LLCs) and proprietorships faced only the tax authority, whose audits were weak
- Ecuador: tax evaded through fictitious purchases was 1.7% for corporations and 11.5% for sole proprietorships (Carrillo et al. 2022)
  - If corporations nonetheless overreport, my estimates are a lower bound
- Corporations have different productivities than the rest of the firms. But that's OK: the critical assumption is a common technology

Model: firms, taxes and overreporting

- Price-taking firms maximize expected after-tax profits; they share a common technology but differ in productivity

$$Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it})$$

  - ω_{it}: productivity, known to the firm; ε_{it}: measurement error in output
- Materials are flexible and deductible; firms report $M_{it}^{*} = M_{it} + e_{it}$, $e_{it} \geq 0$
  - Overreporting is detected with a probability $q(e)$ and costs $\kappa(e, \omega)$; both are zero for a firm that does not overreport

Model: the materials FOC

- The FOC for materials gives the log share of materials in revenue, net of sales taxes:

$$\ln\left(\frac{(1-\tau_P)\rho_t M_{it}}{(1-\tau_S)P_t Y_{it}}\right) = \ln\beta - \varepsilon_{it}$$

  - Productivity does not appear in the share (Hicks-neutral productivity)
- With reported materials M^*_{it}, the log share s_{it} satisfies

$$s_{it} = \ln\beta - \varepsilon_{it} + u_{it}, \qquad u_{it} = \ln(M^*_{it}/M_{it}) \geq 0$$

Testing for overreporting

- Corporations do not overreport: $s_{it} = \ln \beta - \varepsilon_{it}$ gives $\ln \beta = E[s_{it} \mid \text{corporations}]$
- Unincorporated firms: the observed residual $\mathcal{V}_{it} = s_{it} - \ln \beta = u_{it} - \varepsilon_{it}$

$$H_0 : \mu_{\mathcal{V}} = 0 \qquad \text{vs.} \qquad H_1 : \mu_{\mathcal{V}} > 0$$

- One-sided: no firm has an incentive to underreport deductible inputs. A rejection can only point to overreporting
- The test does not depend on the functional forms of the detection probability and the cost of evasion, nor on the Cobb-Douglas form of the technology
- Confidence set for $\mu_{\mathcal{V}}$ by test inversion, clustered by plant

Testing results

Industry	$\hat{\mu}$	95% region	Industry	$\hat{\mu}$	95% region
311 Food products	0.024	$(\cdot, 0.048]$	**342 Printing and publishing**	0.074	[0.036, 0.112]
312 Other food products	−0.018	$(\cdot, 0.076]$	**351 Industrial chemicals**	0.165	[0.001, 0.329]
313 Beverages	0.390	[0.310, 0.470]	**352 Other chemicals**	0.057	[0.002, 0.111]
321 Textiles	0.151	[0.115, 0.188]	356 Plastic products	−0.107	†
322 Wearing apparel	0.224	[0.195, 0.253]	**369 Non-metallic minerals**	0.193	[0.109, 0.277]
323 Leather products	−0.192	†	381 Metal products	−0.005	$(\cdot, 0.025]$
324 Footwear	0.095	[0.059, 0.131]	382 Non-electrical machinery	−0.014	$(\cdot, 0.026]$
331 Wood products	0.250	[0.175, 0.325]	383 Electrical machinery	0.034	$(\cdot, 0.089]$
332 Furniture	−0.007	$(\cdot, 0.059]$	384 Transport equipment	−0.129	†
341 Paper products	−0.066	$(\cdot, 0.004]$	390 Other manufacturing	0.057	$(\cdot, 0.115]$

$\hat{\mu}$: mean of $\mathcal{V}_{it}$ among unincorporated firms. 95% test-inversion region (sharp test); bold: the region excludes 0 (overreporting detected). $(\cdot, b]$: 0 not rejected. †: empty region.

- The test detects overreporting in 9 of the 18 industries liable for the sales tax: 4 of the 5 largest by output; 41% of manufacturing output
- No overreporting in either exempt industry, Food products (311) and Other food products (312): this is expected
- In the other nine liable industries, the test fails to reject: a failure to reject is inconclusive

Appendix

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